Proposal

Automated Ranking and Candidate Interaction System Using Context-Based Job Filtering

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Table of Contents

Table of Contents **2**
I. Background of the Problem **3**
II. Objectives and Target Users **5**
III. Solution Concept **7**
IV. Impact and Benefit Analysis **11**
1. Expected Positive Impacts 11
a. Social Impact 11
b. Economic Impact 11
c. Technological Impact 11
2. Specific Benefits for Target Users 12
a. Benefits for Industry / Companies 12
b. Benefits for Job Seekers 12
c. Benefits for Society 12
V. Conclusion **13**

I. Background of the Problem

Unemployment remains a significant socio-economic issue in Indonesia. According to (Badan Pusat Statistik/BPS), in **August 2024**, the number of unemployed people reached approximately **7.47 million**, with an **Open Unemployment Rate of 4.91%**. A large proportion of this number comes from individuals of productive age, including vocational school graduates and university graduates, indicating a gap between labor supply and industry demand. On the other hand, the Indonesian government has projected the creation of new job opportunities through investments in manufacturing, digital industries, green energy, and infrastructure development. The Ministry of Manpower estimated that **more than 19 million new jobs** would be created during the **2024–2025 period**. However, the availability of job vacancies does not automatically reduce unemployment, as companies face major challenges in the recruitment and selection process, particularly during the initial screening stage.

In practice, the number of job applications received for a single vacancy has increased significantly due to the widespread use of online recruitment platforms. LinkedIn reports that **6,060 job applications are submitted every minute**, which is equivalent to **363,600 applications per hour** and approximately **8.72 million applications per day**. In addition, **61 million users search for job opportunities on LinkedIn every week**, indicating the massive scale of job-seeking activity on digital platforms. This high volume of applications creates a substantial challenge for Human Resources (HR) teams, as they are required to screen and evaluate a large number of resumes within a limited time frame, increasing the complexity and workload of the recruitment process.

The recruitment process has become more complex with the widespread use of **Applicant Tracking Systems (ATS)**, which mainly rely on **keyword-based filtering**. Job applicants are encouraged to include as many relevant keywords as possible—such as skills, tools, job fields, and technologies—to pass the ATS screening. As a result, resumes tend to be longer, repetitive, and not always representative of the applicant’s actual competence.

This condition also negatively affects HR professionals, who are required to read numerous resumes with similar content, increasing the risk of fatigue, oversight, and human error.

A survey conducted by the **Society for Human Resource Management (SHRM)** reported that **67% of recruiters experience burnout during resume screening**, and more than **50% admitted missing qualified candidates** due to time constraints and excessive application volumes. Furthermore, a report by the **IBM Smarter Workforce Institute** stated that inefficient recruitment processes could increase **time-to-hire by up to 40%**, which may reduce organizational productivity and competitiveness.

Based on these challenges, there is a need for a recruitment software solution that goes beyond simple keyword matching and is capable of understanding the **context of applicants' experiences**. Factors such as **years of experience, educational background, relevance of job fields, and relationships between tools and technologies** should be considered holistically. By applying a **context-based filtering approach**, the system is expected to assist HR teams in ranking candidates more accurately and efficiently, reducing manual workload, and improving the quality of recruitment decisions.

https://www.bps.go.id/id/publication/2024/12/09/6f1fd1036968c8a28e4cfe26/keadaan-angkatan-kerja-di-indonesia-agustus-2024.html
https://www.cnnindonesia.com/ekonomi/20251020151602-92-1286510/menaker-bicara-janji-19-juta-lapangan-kerja-di-1-tahun-prabowo-gibran
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https://www.shrm.org/topics-tools/news/talent-acquisition/many-recruiters-path-to-burnout
https://www.ibm.com/new/product-blog/empower-your-organizations-to-make-smart-workforce-decisions

II. Objectives and Target Users

The main objective of this research is to develop a recruitment software that can **simplify and accelerate the candidate screening process**, which is often repetitive, time-consuming, and mentally exhausting for Human Resources (HR) teams. The system aims to reduce the risk of **qualified candidates being overlooked** due to keyword mismatches, non-ATS-friendly resumes, or unconventional experience representations by applying a **context-based filtering approach** rather than relying solely on keyword matching.

This research is conducted with the following specific objectives:

1. To design a candidate screening mechanism that evaluates applicants based on the **context of their experience**, including years of experience, educational background, job field relevance, and tools used.
2. To minimize recruiter fatigue and manual workload during the early recruitment stages by providing an **automated candidate ranking system** that prioritizes relevance over keyword density.
3. To reduce the probability of **false negatives**, where suitable candidates are excluded due to the absence of exact keywords or the use of non-ATS-formatted resumes.
4. To support HR teams in focusing more on **in-depth candidate evaluation**, such as interviews and competency assessment, rather than spending excessive time on initial resume screening.
5. To provide a scalable recruitment solution that can adapt to the **increasing number of job applicants and job vacancies** across various industries.

The target users of this system include **organizations and HR professionals across all industries**, as recruitment is a universal process required in every business sector. As hiring trends continue to grow and applicant volumes increase, HR workloads in resume screening expand faster than the available time and resources for meaningful candidate evaluation. Currently, HR professionals are often required to screen large numbers of resumes within limited timeframes, leaving minimal opportunity to thoroughly understand each candidate's suitability. As a result, the recruitment process becomes focused on **filtering speed rather than candidate quality**.

Furthermore, the rapid growth of service-based industries and digital platforms has led to the emergence of **new job roles with highly specific and evolving responsibilities**. Many of these roles involve tools, workflows, or skill combinations that are still in a **trial-and-error phase** and may not yet be standardized or widely recognized. In such cases, traditional keyword-based filtering becomes ineffective, as relevant experience may not be represented by commonly used terms. Therefore, this system targets all organizations that require a more **flexible, context-aware recruitment approach**, enabling HR teams to identify suitable candidates even when job requirements are new, evolving, or not yet clearly defined in conventional keyword taxonomies.

III. Solution Concept

This proposal presents an AI-based recruitment system designed to simplify and accelerate the candidate screening process, which is often repetitive, time-consuming, and prone to human error. The system aims to **assist** HR teams in identifying suitable candidates more accurately, including qualified applicants who are frequently overlooked due to keyword mismatches or non-ATS-formatted resumes. The recruitment process begins when applicants complete an online application form developed using React and select a job title. The submitted data is sent to a FastAPI backend, where applicant information is structured and stored in a PostgreSQL database. The system then retrieves the corresponding job title and job description from the database and performs AI-based analysis to evaluate candidate suitability.

Using a Large Language Model (LLM) orchestrated through LangChain, the system conducts multi-step contextual reasoning to compare applicant profiles with job requirements. The analysis evaluates multiple variables, including technical tools, frameworks, job field relevance, and work experience duration. Based on this reasoning process, the system generates a compatibility score representing the overall match between the applicant and the position. This score is stored alongside the applicant's data and used to automatically rank candidates for each job position.

HR Dashboard and Interface Design

The system interface is designed as a multi-page yet streamlined HR Dashboard built with React. The dashboard follows a status-driven recruitment workflow while minimizing unnecessary navigation to reduce cognitive load for HR users. The main **Applicants Page** displays a ranked list of candidates based on their compatibility scores. Each candidate initially appears with a default status of **Waiting**, accompanied by a status dropdown. The dashboard also

provides two primary filters: **Job Title** and **Candidate Status**, allowing HR users to quickly refine the displayed candidate list.

From the **Waiting** state, HR users can take action directly through the dropdown by selecting **Interview** or **Reject** without leaving the page. Selecting **Reject** immediately updates the candidate's status in the database and removes the candidate from the active list. Selecting **Interview** updates the candidate's status and redirects them to a dedicated **Interview Page**, while simultaneously triggering workflow automation via n8n for interview scheduling and email notifications.

Interview Management Flow

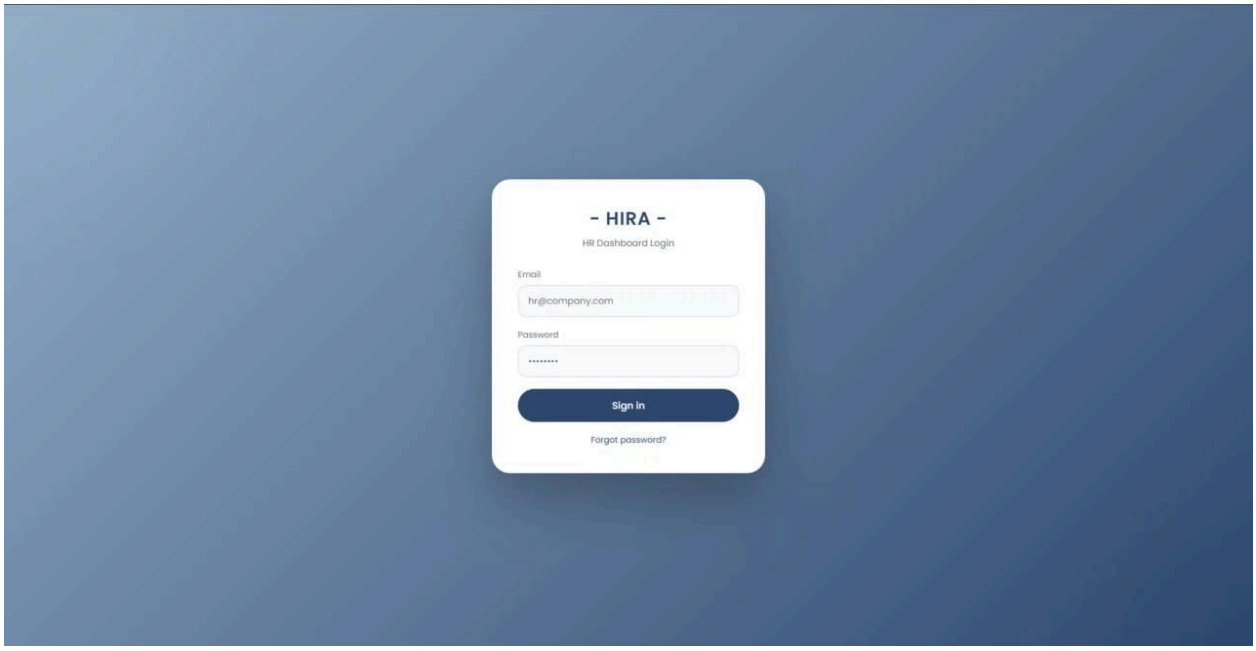
The **Interview Page** is a separate view specifically designed for managing scheduled interviews. This page presents interview-related information, including interviewer details, date and time, meeting link, and interview status. Within this context, the status dropdown is restricted to **Hired** or **Reject**, ensuring logical state progression and preventing invalid transitions. Candidates marked as **Hired** or **Rejected** are no longer displayed in active recruitment views. However, they remain accessible through the Candidate Status filter, allowing HR teams to review recruitment history and decision outcomes within the same dashboard view, without requiring a separate archival interface

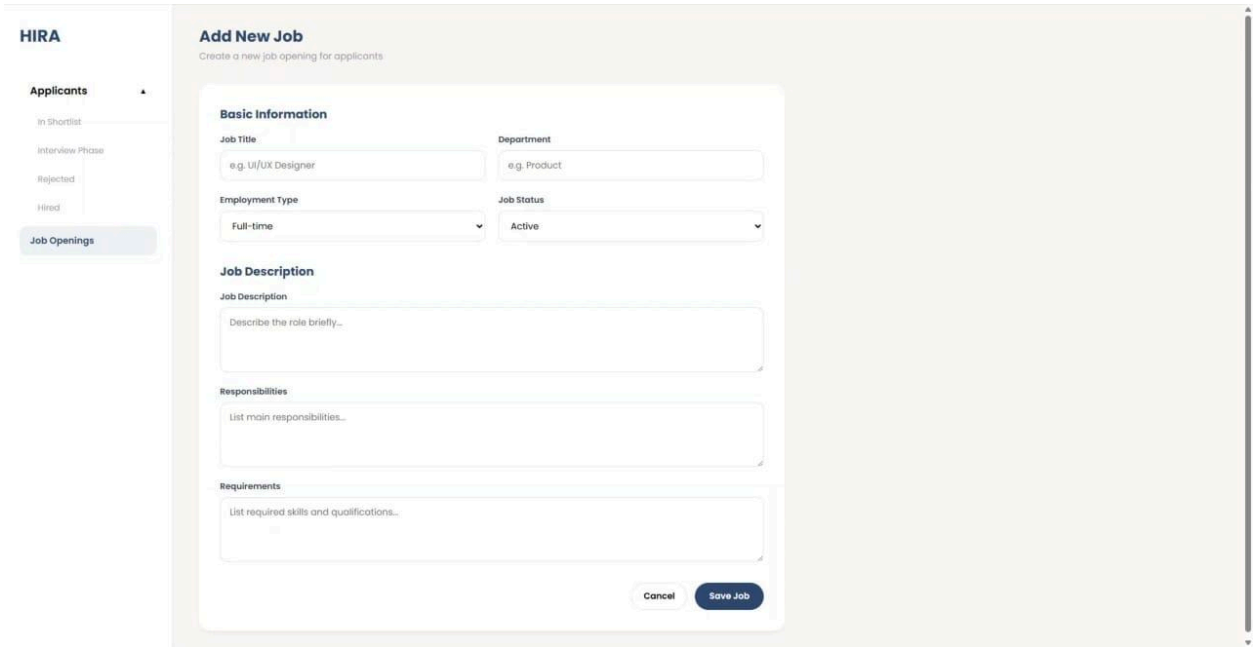
Innovation and Design Rationale

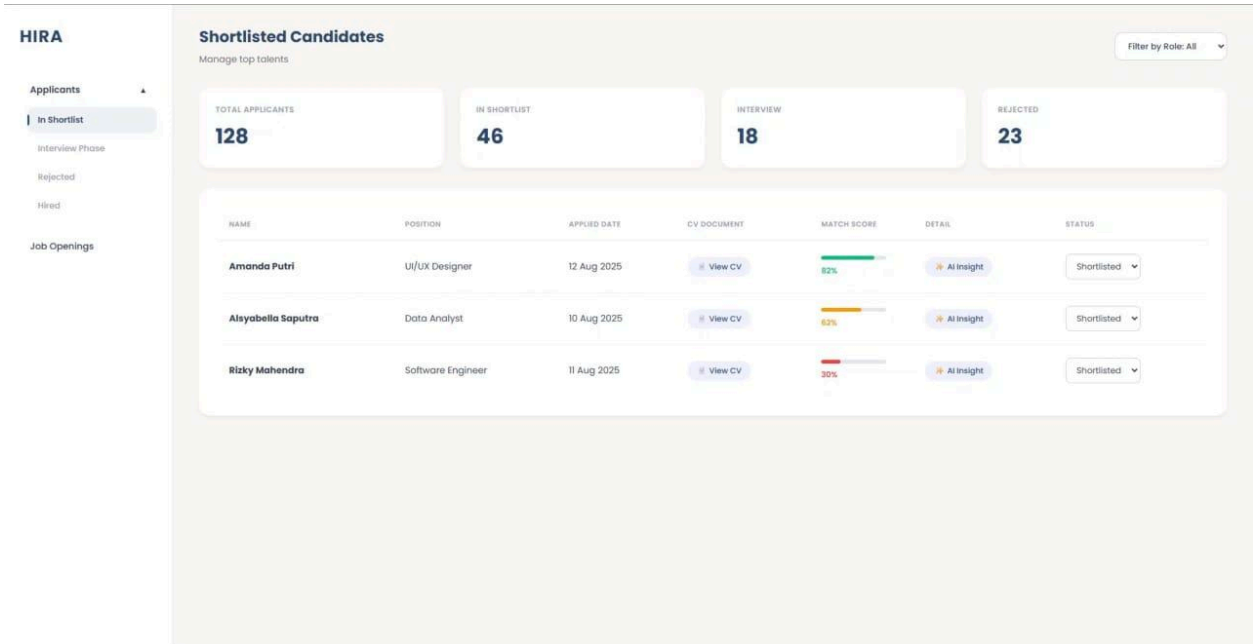
The main innovation of this solution lies in the transition from traditional keyword-based Applicant Tracking Systems (ATS) to a context-based recruitment approach powered by Large Language Models. Instead of relying on exact keyword matching, the system performs semantic and contextual analysis through LangChain-orchestrated reasoning, enabling more accurate, fair, and explainable candidate evaluation—particularly for evolving or non-standard job roles. From a design perspective, the system emphasizes:

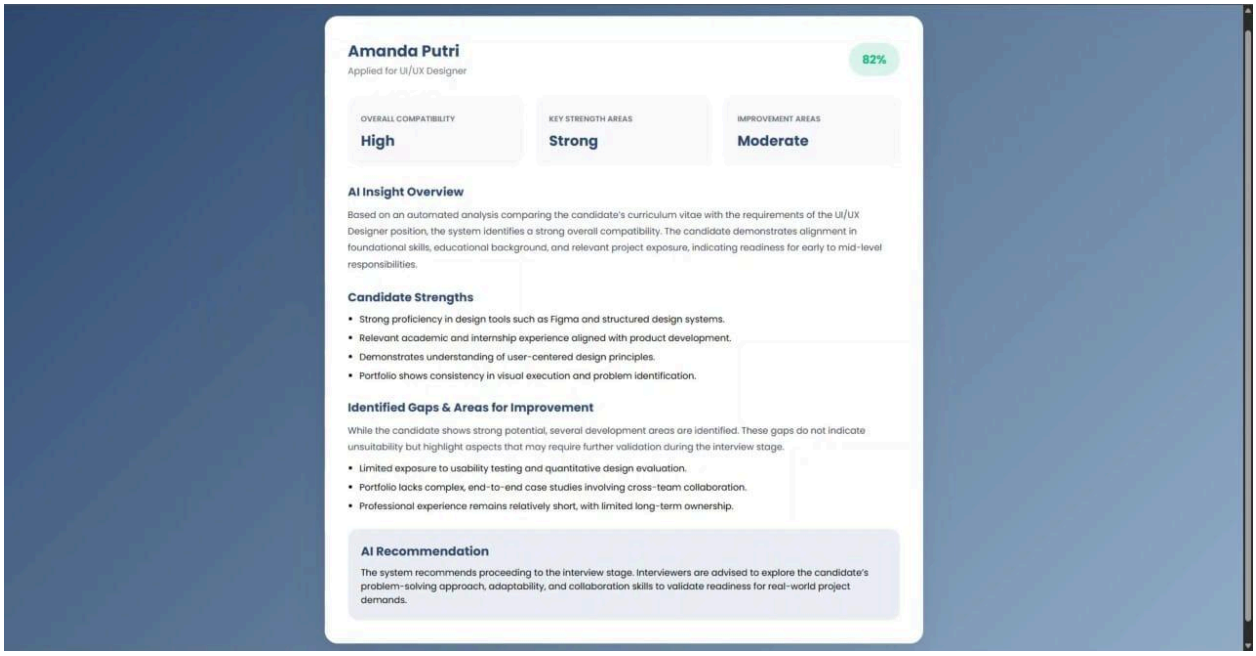
- Clear and enforced state transitions (**Waiting** → **Interview** → **Hired/Reject**)
- Dropdown-based actions to minimize UI complexity
- Page separation only when context changes (general screening versus interview management)

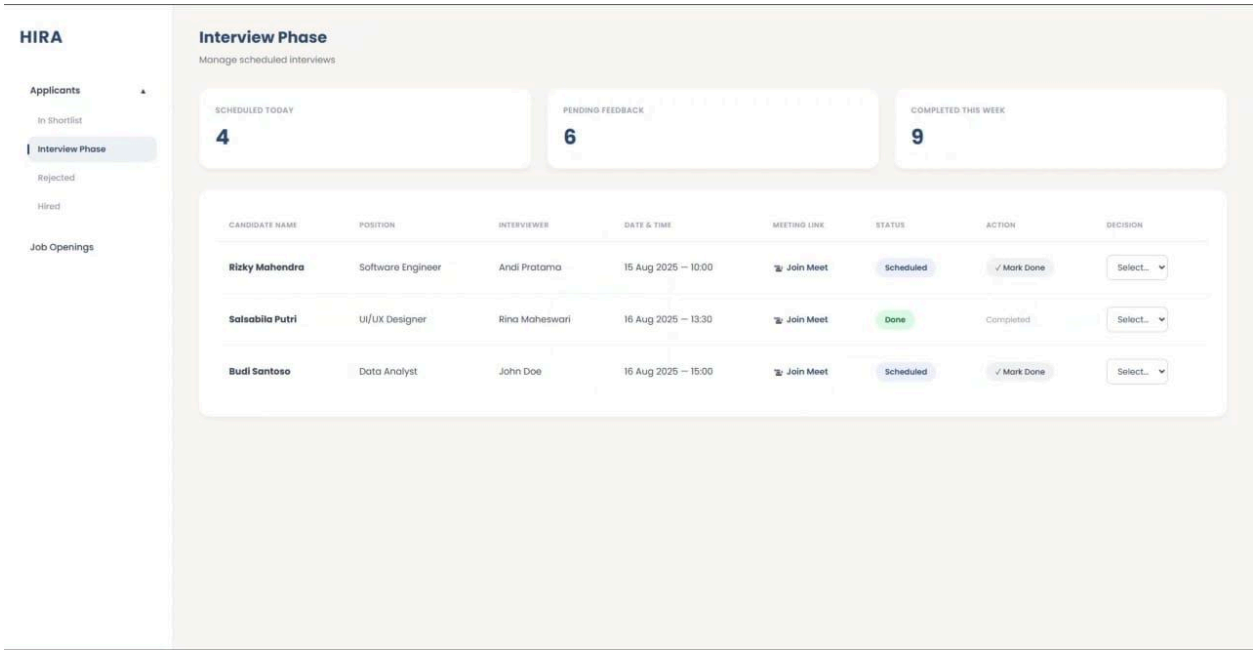
By combining context-aware AI evaluation with dropdown-driven interactions and selective page transitions, the proposed system delivers an efficient, scalable, and easily evaluable end-to-end recruitment prototype. This design makes the solution well-suited for rapid implementation and assessment in hackvidia, academic, or early-stage product evaluation settings.

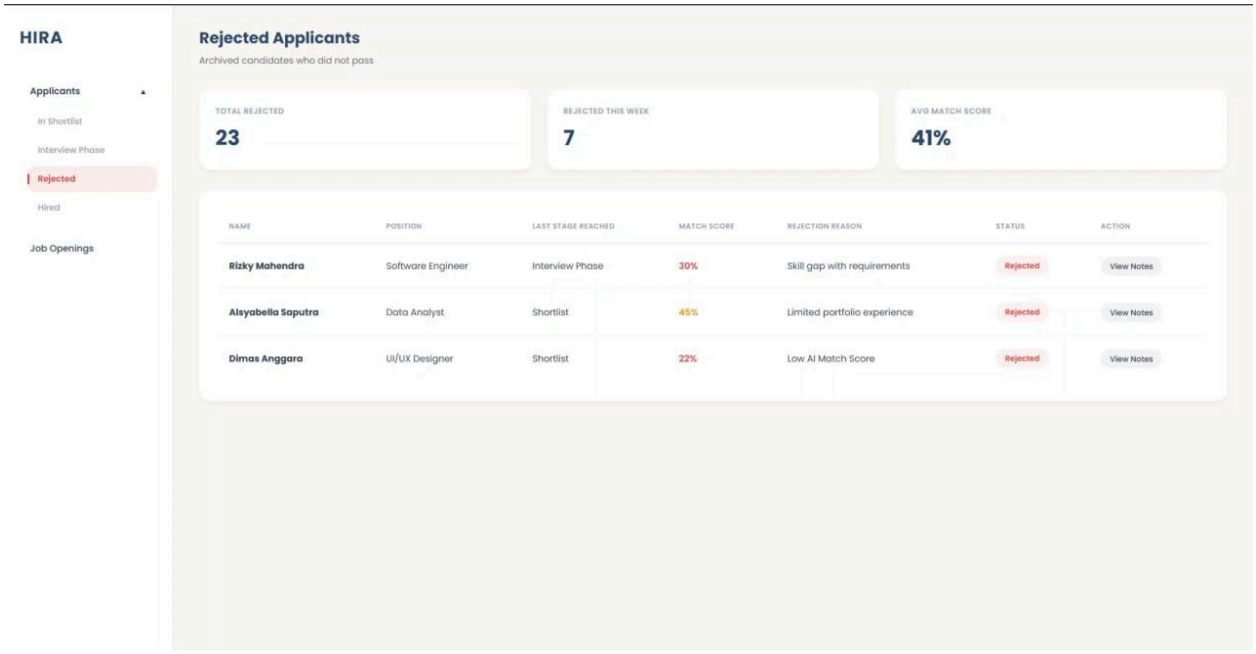

- HIRA -
HR Dashboard Login
Email
hr@company.com
Password
Sign in
Forgot password?

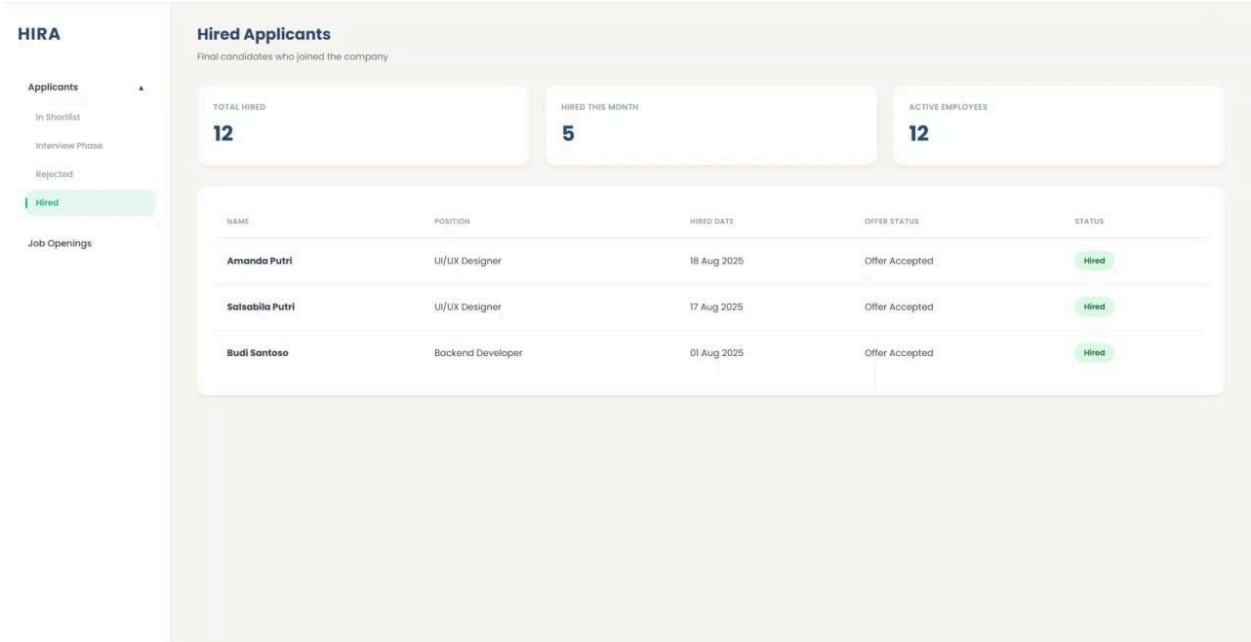

HIRA
Applicants
In Shortlist
Interview Phase
Rejected
Hired
Job Openings
Add New Job
Create a new job opening for applicants
Basic Information
Job Title
e.g. UI/UX Designer
Department
e.g. Product
Employment Type
Full-time
Job Status
Active
Job Description
Job Description
Describe the role briefly...
Responsibilities
List main responsibilities...
Requirements
List required skills and qualifications...
Cancel
Save Job

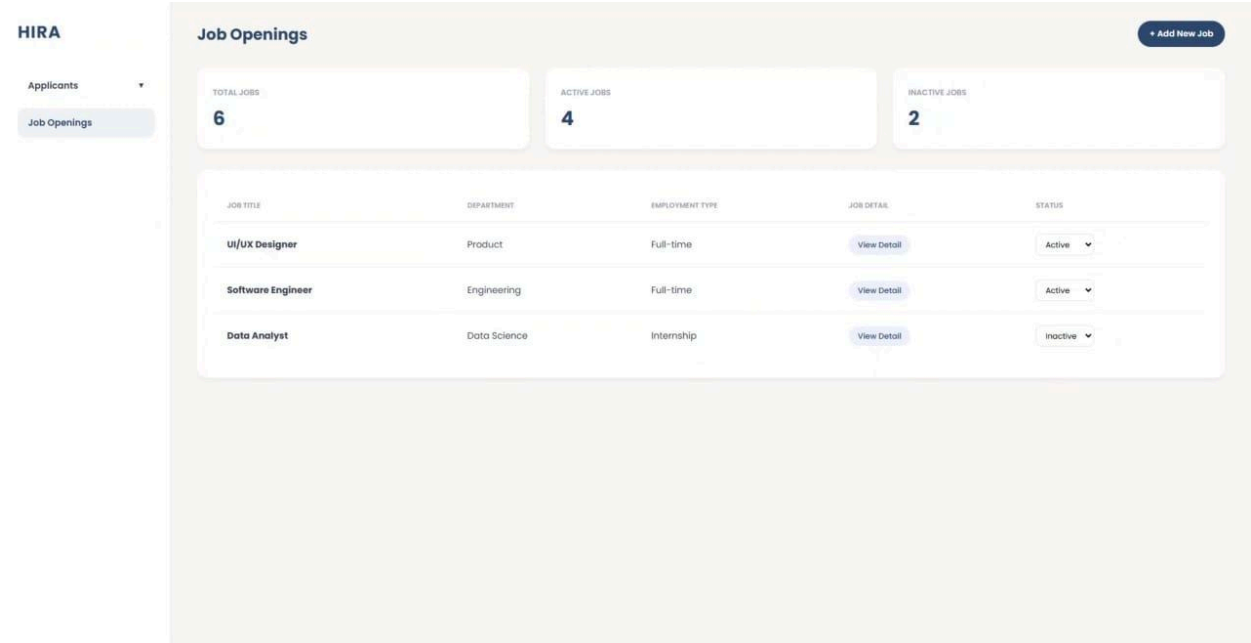

HIRA
Applicants
In Shortlist
Interview Phase
Rejected
Hired
Job Openings
Shortlisted Candidates
Manage top talents
Filter by Role: All
TOTAL APPLICANTS
128
IN SHORTLIST
46
INTERVIEW
18
REJECTED
23
NAME
POSITION
APPLIED DATE
CV DOCUMENT
MATCH SCORE
DETAIL
STATUS
Amanda Putri
UI/UX Designer
12 Aug 2025
View CV
82%
AI Insight
Shortlisted
Alsyabella Saputra
Data Analyst
10 Aug 2025
View CV
62%
AI Insight
Shortlisted
Rizky Mahendra
Software Engineer
11 Aug 2025
View CV
30%
AI Insight
Shortlisted

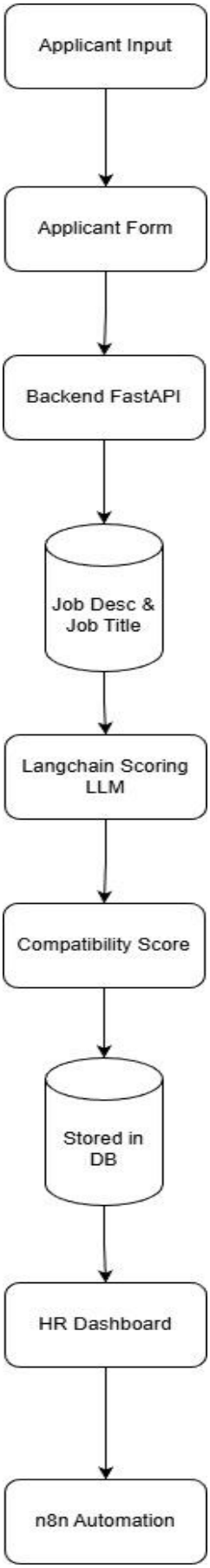

Amanda Putri
Applied for UI/UX Designer
82%
OVERALL COMPATIBILITY
High
KEY STRENGTH AREAS
Strong
IMPROVEMENT AREAS
Moderate
AI Insight Overview
Based on an automated analysis comparing the candidate's curriculum vitae with the requirements of the UI/UX Designer position, the system identifies a strong overall compatibility. The candidate demonstrates alignment in foundational skills, educational background, and relevant project exposure, indicating readiness for early to mid-level responsibilities.
Candidate Strengths
• Strong proficiency in design tools such as Figma and structured design systems.
• Relevant academic and internship experience aligned with product development.
• Demonstrates understanding of user-centered design principles.
• Portfolio shows consistency in visual execution and problem identification.
Identified Gaps & Areas for Improvement
While the candidate shows strong potential, several development areas are identified. These gaps do not indicate unsuitability but highlight aspects that may require further validation during the interview stage.
• Limited exposure to usability testing and quantitative design evaluation.
• Portfolio lacks complex, end-to-end case studies involving cross-team collaboration.
• Professional experience remains relatively short, with limited long-term ownership.
AI Recommendation
The system recommends proceeding to the interview stage. Interviewers are advised to explore the candidate's problem-solving approach, adaptability, and collaboration skills to validate readiness for real-world project demands.


HIRA
Applicants
In Shortlist
Interview Phase
Rejected
Hired
Job Openings
Interview Phase
Manage scheduled interviews
SCHEDULED TODAY
4
PENDING FEEDBACK
6
COMPLETED THIS WEEK
9
CANDIDATE NAME
POSITION
INTERVIEWER
DATE & TIME
MEETING LINK
STATUS
ACTION
DECISION
Rizky Mahendra
Software Engineer
Andi Pratama
15 Aug 2025 — 10:00
Join Meet
Scheduled
✓ Mark Done
Select...
Salsabila Putri
UI/UX Designer
Rina Maheswari
16 Aug 2025 — 13:30
Join Meet
Done
Completed
Select...
Budi Santoso
Data Analyst
John Doe
16 Aug 2025 — 15:00
Join Meet
Scheduled
✓ Mark Done
Select...


HIRA
Applicants
In Shortlist
Interview Phase
Rejected
Hired
Job Openings
Rejected Applicants
Archived candidates who did not pass
TOTAL REJECTED
23
REJECTED THIS WEEK
7
AVG MATCH SCORE
41%
NAME
POSITION
LAST STAGE REACHED
MATCH SCORE
REJECTION REASON
STATUS
ACTION
Rizky Mahendra
Software Engineer
Interview Phase
30%
Skill gap with requirements
Rejected
View Notes
Alsyabella Saputra
Data Analyst
Shortlist
45%
Limited portfolio experience
Rejected
View Notes
Dimas Anggara
UI/UX Designer
Shortlist
22%
Low AI Match Score
Rejected
View Notes


HIRA
Applicants
In Shortlist
Interview Phase
Rejected
Hired
Job Openings
Hired Applicants
Final candidates who joined the company
TOTAL HIRED
12
HIRED THIS MONTH
5
ACTIVE EMPLOYEES
12
NAME
POSITION
HIRED DATE
OFFER STATUS
STATUS
Amanda Putri
UI/UX Designer
18 Aug 2025
Offer Accepted
Hired
Salsabila Putri
UI/UX Designer
17 Aug 2025
Offer Accepted
Hired
Budi Santoso
Backend Developer
01 Aug 2025
Offer Accepted
Hired


HIRA
Applicants
Job Openings
Job Openings
+ Add New Job
TOTAL JOBS
6
ACTIVE JOBS
4
INACTIVE JOBS
2
JOB TITLE
DEPARTMENT
EMPLOYMENT TYPE
JOB DETAIL
STATUS
UI/UX Designer
Product
Full-time
View Detail
Active
Software Engineer
Engineering
Full-time
View Detail
Active
Data Analyst
Data Science
Internship
View Detail
Inactive


Applicant Input
Applicant Form
Backend FastAPI
Job Desc &
Job Title
Langchain Scoring
LLM
Compatibility Score
Stored in
DB
HR Dashboard
n8n Automation

IV. Impact and Benefit Analysis

1. Expected Positive Impacts

a. Social Impact

This solution contributes to a more fair and inclusive recruitment process. By applying a context-based evaluation approach, the system reduces bias caused by differences in CV formats, technical terminology, or non-conventional educational backgrounds. As a result, competent job seekers who are often eliminated by keyword-based Applicant Tracking Systems (ATS) are given broader opportunities to be considered. Additionally, a more transparent and structured recruitment process can increase job seekers' trust in the selection systems used by companies.

b. Economic Impact

From an economic perspective, the proposed system helps organizations reduce recruitment time and costs (time-to-hire). Automation of candidate screening, ranking, and interview scheduling significantly lowers the administrative workload of HR teams and enables more efficient allocation of human resources. By identifying suitable candidates earlier in the recruitment process, companies can also minimize the risk of mis-hiring, which often leads to increased operational costs and reduced long-term productivity.

c. Technological Impact

Technologically, this solution promotes the adoption of Artificial Intelligence in a more practical and contextual manner within Human Resource Management systems. The shift from keyword-based filtering to context-based analysis demonstrates the use of AI not merely as an automation tool, but as a decision support system. Furthermore, the integration of Large Language Models (LLMs) with workflow automation illustrates a modular and scalable application of AI technology that aligns with current industry needs.

2. Specific Benefits for Target Users

a. **Benefits for Industry / Companies**

- Accelerates candidate screening in environments with high application volumes
- Produces more accurate and relevant candidate rankings based on job requirements
- Reduces HR administrative workload through automated interview scheduling and email communication
- Provides structured recruitment history through a status-based system (Waiting, Interview, Reject, Hired)
- Supports data-driven and context-aware decision-making rather than keyword-based filtering

b. **Benefits for Job Seekers**

- Increases opportunities for candidates with relevant competencies, even when different terminology or CV formats are used
- Reduces dependence on keyword optimization to pass ATS screening
- Ensures a more objective selection process aligned with actual skills and experience

c. **Benefits for Society**

- Improves labor market efficiency through more accurate matching between candidates and job roles
- Helps reduce structural unemployment caused by skill and job requirement mismatches
- Encourages digital transformation in recruitment processes that can adapt to technological advancements and emerging job roles

V. Conclusion

The increasing volume of job applications and the limitations of traditional keyword-based recruitment systems have made candidate screening a complex and time-consuming process for HR teams. These challenges often result in qualified candidates being overlooked while increasing the administrative burden on recruiters. The proposed AI-based recruitment system addresses this problem by introducing a context-based evaluation approach that enables more accurate, fair, and efficient candidate screening.

By leveraging Large Language Models (LLMs) for contextual analysis, the system shifts recruitment practices from simple keyword matching toward intelligent decision support. The integration of automated scoring, candidate ranking, and interview workflow automation reduces time-to-hire while allowing HR teams to focus on deeper and more meaningful candidate evaluation. This solution demonstrates how Artificial Intelligence can be applied pragmatically to solve real-world recruitment challenges without increasing system complexity.

Through the development of this solution, the team expects to demonstrate the potential of AI-driven recruitment systems in improving screening accuracy, reducing operational inefficiencies, and promoting inclusive hiring practices. The proposed system is designed not only as a functional prototype but also as a scalable foundation that can be further developed to support more advanced HR decision-making features in the future.

It is expected that this solution can contribute to the digital transformation of recruitment processes, particularly in organizations facing high application volumes and evolving job requirements. Ultimately, the team hopes that this project can serve as a reference for the adoption of context-based AI systems in recruitment and inspire further innovation in the application of Artificial Intelligence within Human Resource Management.